Day 5: Same DNA, different jobs

- Today's target: DNA, through proteins, gives different cells their different jobs (LS1-1.5)

Warm-up

- Your skin cells and your nerve cells have the same DNA.
- Why are they so different?

Key ideas

- Every cell has the same DNA, including all the genes.
- Each kind of cell uses a different set of genes.
- The proteins a cell makes decide its job.
- In a cat, only pigment-making cells use the gene for the pigment-making protein.

Part A: two kinds of cells

Cell	Genes in cell	Proteins made	Job
P	1, 2, 3, 4	1 and 2	Carries signals
Q	1, 2, 3, 4	3 and 4	Squeezes to move the body

Same DNA? Why different jobs?

Part B: cells in a Siamese cat's ear

- Which cells have the gene for the pigment-making protein?
- Which cells make the protein?
- Fill in the table: pigment-making cell, skin cell, nerve cell.

Part C: where does the changed protein work?

- One amino acid is different. The protein works only in skin cooler than about 33 °C (a simplified number).
- Use the temperatures to fill in each body part: protein works? Fur dark or cream?

Body part	Skin temp (°C)
Ear	28
Nose	30
Paw	28
Tail tip	27
Upper leg	34
Back	36
Belly	36

Explain it

- Use the words DNA, gene, protein, and job to explain why a Siamese cat has dark points.
- Then: Exit ticket CFA LS1-1.5 (6 minutes).